10-1 Types of Data and Analysis

Point!

1 Types of Data

(1) ([1]**Quantitative data**) : Data that is expressed in numbers and where the size of the numbers has meaning.

[1] ([2]**Interval scale**) : Data where the intervals between numbers have meaning.
<Examples> Time, temperature, calendar year, etc.

[2] ([3]**Ratio scale**) : Data where not only the intervals between numbers but also the ratios of the numbers have meaning.
<Examples> Length, weight, age, etc.

(3) ([4]**Qualitative data**) : Data used to distinguish classifications or types.

[1] ([5]**Nominal scale**) : Data that only holds meaning in distinguishing different categories.
<Examples> Blood type, prefecture, phone number, etc.

[2] ([6]**Ordinal scale**) : Data in which the order is meaningful but the intervals are not meaningful.
<Examples> Grades on report cards, test rankings, grade level, etc.

3 Data Analysis Approachs

(1) ([7]**Discriptive Statistics**) : Summarizing and organizing historical data to get insights about the characteristics of a sample

[1] ([8]**Measures of centeral tendency**) : Mean, Median,and Mode.

[2] ([9]**Measures of Spread**) : Range, Variance,and Standard Deviation

(2) ([10]**Inferential Statistics**) : Used to make inferences, predictions, and generalizations about the entire population from the data of a sample.

[1] ([11]**Hypothesis Testing**)

[2] ([12]**Confidence Intervals**)

Warm Up

Answer the following questions.

(1) For the data shown in [1] to [6] below, answer A for quantitative data and B for qualitative data.

[1] Time [2] Blood type [3] Names of U.S. states
[4] Weight [5] Test ranking [6] Age

(2) Into which scale of measurement can the data in [1] to [6] be classified? Choose one from the options **A** to **D** below, and answer using the letters.

[1] Time [2] Blood type [3] Names of U.S. states
[4] Weight [5] Test ranking [6] Age

A Interval scale **B** Ratio scale **C** Nominal scale **D** Ordinal scale

Explanation

(1) [1] A [2] B [3] B [4] A [5] B [6] A
(2) [1] **A** [2] **C** [3] **C** [4] **B** [5] **D** [6] **B**

Try

1 Answer the following questions.

(1) For the data shown in [1] to [9] below, answer A for quantitative data and B for qualitative data.

[1] Marathon times [2] License plate numbers [3] Number of cars
[4] Sales revenue of a company [5] Impressions of cooking [6] Days of the week
[7] Length of traffic jams [8] Addresses [9] Favorite foods

(2) Choose the description for the characteristics of scales of measurements [1] to [4] from the options **A** to **D** in [Group A]. Then, choose the item that represents the scale from the options a to d in [Group B]. Answer both questions using the letters.

[1] Interval scale [2] Ordinal scale [3] Ratio scale [4] Nominal scale

[Group A]

A A scale with equal intervals but without meaningful ratios
B A scale where the order has meaning, but the intervals do not
C A scale represented by names or characteristics for classification and differentiation
D A scale where the origin is defined and both intervals and ratios have meaning

[Group B]

a Grade b Product prices c Phone numbers d Western calendar years

Exercise

1 Answer the following questions.

(1) Choose the correct explanation for nominal data from the options **A** to **D** below, and answer using the letter.

A Data that is expressed in numbers and where the size of the numbers has meaning.

B Data published on the internet by governments or businesses that is freely accessible and usable by businesses.

C Data to which numbers are assigned that have a meaningful order, but where the intervals have no meaning.

D Data represented by names or characteristics for distinguishing and classifying subjects.

2 Answer the following questions.

(1) Choose one scale of measurement for each of the data types shown in [1] to [6] below from the options **A** to **D** below, and answer using the letters.

[1] Temperature

[2] Occupation

[3] Number of visitors to an amusement park

[4] Uniform numbers of baseball players

[5] Number of cars sold

[6] Marathon ranking

A Interval scale **B** Ratio scale **C** Nominal scale **D** Ordinal scale